From Chaos To Coalescent Parsimony

Deriving the Inverse-Square Law, the Baryon-to-Photon Ratio, and the Constants of Electromagnetism. The Mechanics Behind Gravity.

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Abstract

We present a framework in which gravity is not a fundamental interaction but emerges from the collective interference of coherent electromagnetic instruction carriers that structure spacetime kinematics. By modeling gravitational attraction as a consequence of conserved radial flux in a coherent field, we derive the classical $1/r^2$ force law without invoking the equivalence principle or curvature as primary. A coherence factor $C = 1 + R_b/3$ naturally appears, parameterizing deviations from classical gravity due to the finite coherence length of the carrier field, where R_b approximately equals 0.26 is the ratio of baryon-to-photon energy densities at matter-radiation equality (derived in Paper 15 [3], Section 3.1, from the collision-cascade geometry of the coherence background). The variational foundation for the modified Einstein equations and the NIPOK metric is established in Paper 14 [2]; the derivations of $A^* = 5.970$, the coherence length xi, and R_b are completed in Paper 15 [3]. The observational confirmation of the predicted coherence modifications to gravitational dynamics across galactic and cluster scales is presented in Paper 16 [1]. The present work (Paper 17) establishes the foundational derivation of the inverse-square law, traces the logical chain connecting the coherence factor to cosmological parameters, and provides a roadmap to the constants of electromagnetism from first principles. All quantities that are cited rather than derived here are clearly flagged as such.

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1. Fundamental Ontology

We define the ontological primitives upon which the emergent theory of gravitational dynamics is constructed. In this framework, spacetime is not a fundamental entity but a derived construct arising from the collective behavior of a coherent electromagnetic substrate. The variational foundation for this ontology is established in Paper 14 [2], where the Successive Collision Theory (SCT) action S_SCT is shown to recover the SCT-MASTER equation and the NIPOK (Named Invariant Polar-Oblate Kinematic) metric as on-shell solutions.

(a) Instruction Carriers

The foundational objects are instruction carriers, defined as propagating degrees of freedom in the electromagnetic field that encode phase information. Each carrier is characterized by a wavefunction of the form

$$\psi(x,t) = A \exp(i\phi(x,t))$$

where A is an amplitude and $\phi(x,t)$ is a coherence phase that encodes directional and relational information between localized sources. These carriers are not merely radiative modes but constitute a structuring field that mediates the effective configuration of what is conventionally interpreted as spacetime geometry.

The dynamics of phi are governed by a conserved current condition

$$\partial_\mu J^\mu = 0, \quad J^\mu = \rho \, \partial^\mu \phi,$$

where rho is the instruction carrier density. This conservation enforces radial flux constancy in static configurations, directly yielding the 1/r squared force law as a geometric consequence of flux conservation in a coherent medium.

(b) Spheres of Influence

An assembly of instruction carriers forms a sphere of influence when the local coherence exceeds a critical threshold. Let the coherence function C(x,t) satisfy $C(x,t) \geq C_{crit}$, where C_{crit} is a universal constant marking the transition from incoherent electromagnetic fluctuations to a phase-locked collective state. Within a sphere of influence, the phases phi become correlated over macroscopic scales, and the system exhibits emergent effective geometry. The boundary of a sphere of influence is not a material surface but a coherence isosurface. Matter. understood as stable phase-locked configurations of instruction carriers, manifests as localized regions of elevated coherence, each surrounded by a gradient in C-hat that determines the extent of its gravitational influence.

(c) Coherence Function C(x,t) and Coupling Enhancement Factor C-hat

The coherence function is defined operationally as the ratio of coherent intensity to total incoherent intensity among the ensemble of N instruction carriers contributing at (x, t):

$$\hat{C}(x,t) = 1 + |\Psi_{source}(x,t)|^2 / I_{bg}(x,t)$$

where I_{bg} is the total incoherent background intensity that would be present at the test point even in the absence of the local source (the cosmological noise floor), and Ψ_{source} is the coherent field from the local source mass alone. The "1" represents the background contribution to the total intensity; the second term $|\Psi_{source}|^2/I_{bg}$ is the source enhancement above that background. Because $I_{bg} > 0$ always, $\hat{C} \geq 1$ always; a strong source gives $\hat{C} \gg 1$. This form naturally produces $\hat{C}_{bg} = 1 + R_b/3$ when $|\Psi_{source}|^2/I_{bg} = R_b/3$ is evaluated for the cosmological background (Paper 15 [3] Section 3.2). In fully virialized halos C-hat reaches a maximum $A^* = 5.970$. Throughout this paper the symbol C without a hat denotes the microscopic coherence function c (bounded by 1), and C-hat (or C_{bg}, C(r)) denotes the gravitational coupling enhancement factor. The

detailed derivation of I_bg and the resulting value of C-hat_bg is the content of Paper 15 [3], Section 3.

Definition of "source" and "background" in the general case. The split Psi = Psi_source + Psi_background is operationally defined as follows: when computing the gravitational force exerted by mass M_1 on a test point P, Psi_source is the coherent field produced by M_1 at P, and I_bg is the total incoherent intensity at P from all other matter (i.e., from the rest of the universe excluding M_1). This split is well-defined for an isolated two-body system. For an N-body system (e.g., a galaxy cluster), the force on a test mass M_2 from each other mass M_j is computed separately using C-hat_j = 1 + |Psi_j|^2/I_bg,j, where Psi_j is the field from M_j alone and I_bg,j is the incoherent background from all sources other than M_j. The total force is the sum over all j. This pairwise decomposition is observer-independent in the sense that the identity of "source" is tied to the mass generating the force, not to the observer. In situations where the source is an extended mass distribution rather than a point mass, Psi_source is the coherent superposition from all elements of that distribution, while I_bg excludes those elements. The coherence of extended sources is governed by the continuum integral of Section 2.4. We acknowledge that the pairwise decomposition introduces an approximation when sources are close enough for their carrier fields to interfere with each other. The exact treatment requires the full N-body coherence field equations of Paper 14 [2], and is an active area of development.

Crucially, the effective gravitational coupling strength becomes a function of C-hat. In the emergent theory, the Newtonian potential is modulated as

```
Phi_eff(x) = Phi_Newton(x) * C-hat(x),
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with deviations from classical gravity arising wherever C-hat is not equal to 1. In the cosmological weak-field limit C-hat evaluates to 1 + $R_b/3$, where R_b is the baryon-to-photon energy density ratio at matter-radiation equality. The derivation of this value is completed in Paper 15 [3]; Section 3.5 of this paper summarizes the logical chain and states the result.

(d) Role of the Metric

The effective metric is a derived quantity, not a fundamental field. It is constructed from the instruction carrier phase field and the coherence function. Its role is therefore epistemological rather than ontological: it is a convenient mathematical encoding of the phase ordering present in the underlying instruction carrier field. Curvature in this effective metric is not a primitive but a manifestation of gradients in the coherence field. In regions where C is constant and maximal, the effective metric reduces to flat Minkowski. In regions where coherence gradients exist, the metric acquires effective curvature that reproduces the phenomenology of general relativity in the classical limit.

Summary of Ontology

Ontological primitive	Status
Spacetime	Derived (coarse-grained phase ordering)

Instruction carriers	Fundamental (EM-based, phase-structured)
Coherence function C(x,t)	Fundamental (governs emergent coupling)
Effective metric	Derived (describes phase-ordered geometry)
Gravity	Emergent (coherence-gradient force, not fundamental)

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2. Coherence Function

2.1 Definition and Microscopic Formulation

Consider an ensemble of instruction carriers, each described by a wavefunction

$$\psi_i(x,t) = A_i(x,t) \exp(i\phi_i(x,t)),$$

where A_i is the amplitude and ϕ_i the phase of the i-th carrier. The total field at position x and time t is the coherent sum

$$\Psi(x,t) = \sum_i \psi_i(x,t) = \sum_i A_i \exp(i\phi_i).$$

The microscopic coherence function is defined as the normalized coherent intensity among the local ensemble of N carriers:

$$c(x,t) = |\Psi_{local}(x,t)|^2 / (\sum_i |A_i(x,t)|^2), \quad c \text{ in } [0, 1].$$

This definition satisfies $0 \le c \le 1$ exactly. The numerator $|\Psi_{local}|^2$ represents the coherent intensity from local carriers; the denominator $\sum|A_i|^2$ is their total incoherent (power-sum) intensity. Expanding the numerator:

$$|\Psi_{local}|^2 = \sum_i |A_i|^2 + 2\sum_{i<j} A_i A_j \cos(\phi_i - \phi_j),$$

so the coherence function becomes

$$c(x,t) = 1 + [2\sum_{i<j} A_i A_j \cos(\Delta\phi_{ij})] / [\sum_i |A_i|^2],$$

where $\Delta\phi_{ij} = \phi_i - \phi_j$. The cross-term sum is bounded above by $[(\sum|A_i|)^2 - \sum|A_i|^2]/2$ (Cauchy-Schwarz), so $c \le 1$ exactly; $c = 1$ only when all phases are identical.

Gravitational coupling enhancement factor C-hat. For the purposes of the gravitational force law, what matters is not c but the ratio of source-coherent intensity to the cosmological background incoherent intensity I_{bg} at that point. We define the gravitational coupling enhancement factor:

$$\hat{C}(x,t) = 1 + |\Psi_{source}(x,t)|^2 / I_{bg}(x,t).$$

Here I_{bg} is the total incoherent background intensity from all other matter in the universe. Since $I_{bg} > 0$ always, $\hat{C} \ge 1$ always, with equality only when the source

intensity is zero. A strong source gives C-hat >> 1. In the cosmological weak-field limit, Paper 15 [3] Section 3.2 evaluates |Psi_source|^2/I_bg = R_b/3 and derives C-hat_bg = 1 + R_b/3 approximately equals 1.087; in fully virialized halos C-hat reaches a maximum A* = 5.970 [3]. The symbols C_bg, C(r), and C-hat throughout this paper always refer to this enhancement factor, not to the microscopic function c. The two quantities are equal in the single-source, no-background limit (I_bg approaches Sum|A_i|^2), but differ in the cosmological context.

2.2 Amplitude Decay with Distance

Each instruction carrier originates from a source and propagates outward. For a carrier emitted from a point source at position r-prime, conservation of flux in three-dimensional space imposes an amplitude decay

```
A_i(x) proportional to 1 / |x - r_i-prime|.
```

This inverse-distance scaling follows from the conserved current condition partial_mu J^mu = 0 with J^mu = rho partial^mu phi. In a static, spherically symmetric configuration, the radial flux 4*pi*r^2*rho*v_r is constant. We additionally assume that the radial propagation speed v_r is independent of r (carriers propagate radially at a fixed speed, consistent with a linear dispersion relation omega = v_r k). Under this assumption rho is proportional to 1/r^2, and since amplitude A is proportional to sqrt(rho) (wave intensity A^2 is proportional to carrier density rho times a constant v_r), we obtain A proportional to 1/r. If v_r were r-dependent, the decay law would be modified accordingly; the r-independence of v_r is inherited from the carrier dispersion relation established in Paper 14 [2]. A minor tension exists: Section 2.3 introduces a spectral width Gamma, implying that carriers are not perfectly monochromatic, which in a dispersive medium could produce r-dependent group velocities. This paper assumes that any dispersion is negligible over astrophysical scales of interest. Formally, |dv_r/dk| * Delta_k * (r/v_r) << 1. Verifying this condition requires the explicit dispersion relation from Paper 14 [2], and its violation would modify the A proportional to 1/r scaling at large r.

2.3 Temporal Correlation Function

The phases phi_i evolve in time. For a carrier propagating from source to observation point, the phase accumulated along the path is

```
phi_i(x,t) = phi_i^(0) + omega_i t - k_i*(x - r_i-prime),
```

where omega_i and k_i satisfy a dispersion relation. The temporal correlation function is

```
f(Delta_t) = <exp(i*Delta_phi_ij(t+Delta_t)) exp(-i*Delta_phi_ij(t))>
 = integral P(Delta_omega) exp(i*Delta_omega*Delta_t) d(Delta_omega),
```

with P(Delta_omega) the spectral distribution of frequency differences. For a distribution of width Gamma, the correlation decays as f(Delta_t) ~ exp(-Gamma|Delta_t|). The coherence time tau_c = 1/Gamma sets the timescale over which phase correlations persist.

2.4 Continuum Limit and Integral Formulation

For a continuous distribution of sources with density rho_s(r-prime), the total field becomes

```
Psi(x,t) = integral [rho_s(r-prime) / |x - r-prime|] exp(i*phi(r-
                    prime,x,t)) d^3r-prime.
```

The coherence function in the continuum limit is then

```
c(x,t) = |Psi_local(x,t)|^2 / integral [rho_s(r-prime)^2 / |x - r-
                    prime|^2] d^3r-prime.
```

The denominator integral rho_s(r-prime)^2 / |x - r-prime|^2 d^3r-prime is the total incoherent intensity in the continuum limit, obtained by squaring and integrating the amplitude distribution without cross-interference terms. This quantity is well-defined for physically reasonable (square-integrable) source distributions and serves as the normalization that keeps c <= 1. Here c denotes the microscopic coherence function (Section 2.1): all carriers contributing at (x, t) form the local ensemble, and the denominator is their total incoherent power sum. This is distinct from the gravitational coupling enhancement factor C-hat = 1 + |Psi_source|^2/I_bg defined in Section 1(c), which compares source intensity to the cosmological background intensity and can exceed 1.

2.5 Coherence Length and Phase Diffusion

The dynamics of the microscopic coherence function c (Section 2.1, c in [0,1]) are governed by a phase-diffusion-relaxation equation. Let v_phi = nabla phi/m be the carrier velocity field, where m is an effective inertia scale with dimensions of mass that sets the ratio of phase gradient to carrier velocity. Note that m does not appear in the equilibrium coherence length xi = sqrt(D/Gamma_c): it enters only through the product D = D-tilde/m^2, where D-tilde is a phase diffusivity with units [length^2 * mass/time], so m cancels in xi and is not an independent free parameter of the theory. The evolution of c obeys

```
partial_t c + nabla*(c v_phi) = D nabla^2 c + Gamma_c(1 - c),
```

where D is the phase-diffusion coefficient and Gamma_c is the coherence relaxation rate (distinct from the fine-structure constant alpha used in the Bonus section below). This equation governs the microscopic coherence function c: the relaxation term Gamma_c(1 - c) drives c toward its maximum value of 1, which is physically appropriate since c measures local phase alignment and full alignment (c = 1) is the attractor. The equilibrium coherence length xi is set by the balance between diffusion (which tends to reduce spatial coherence) and the relaxation source term (which drives c toward 1). Setting partial_t c = 0 and seeking a spatially decaying solution gives xi = sqrt(D/Gamma_c). Note that without the diffusion term D nabla^2 c the equation would have no intrinsic length scale; the diffusion term is essential and is included here. The gravitational coupling enhancement factor C-hat = 1 + |Psi_source|^2/I_bg is a derived quantity: once c (and therefore the coherent source field Psi_source) is determined by this equation, C-hat follows from evaluating |Psi_source|^2 relative to the cosmological background intensity I_bg. A separate dynamical equation for C-hat is not required; it is fully determined by c and the cosmological background, whose

evaluation is in Paper 15 [3]. The numerical value of xi for the cosmological case is derived in [3].

2.6 Coherence Decay and Screening

In regions distant from sources, the phase factor exp(ik|x - r-prime|) oscillates rapidly as a function of r-prime, causing the numerator integral to average toward zero unless the source distribution is correlated with the phase. This yields a decay of coherence with distance from matter concentrations on scales exceeding xi.

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3. Derivation of the Inverse-Square Law

We present a first-principles derivation of the inverse-square force law from the coherence properties of instruction carriers. The derivation does not assume the inverse-square form a priori; it derives it as a consequence of conserved radial flux in a coherent field. Modifications from finite coherence length are encoded in the factor (1 + R_b/3), whose derivation is deferred to [3] as stated explicitly in Section 3.5. The argument proceeds in six steps.

3.1 Setup: Point Mass and Isotropic Emission

Consider a point mass M_1 at rest at the origin. This mass corresponds to a stable, phase-locked configuration of instruction carriers that emits carriers isotropically. Let the source strength be proportional to M_1:

```
dN/dt = kappa * M_1,
```

where kappa is a coupling constant. Because the emission is isotropic, the flux of carriers through a sphere of radius r is

```
Phi_carriers(r) = kappa M_1 / (4*pi*r^2).
```

No assumption about the force law has been made at this stage.

3.2 Flux Conservation and Amplitude Decay

Each instruction carrier is described by a wavefunction psi = A exp(i*phi). Conservation of the carrier current J^mu = rho partial^mu phi implies nabla*(rho v) = 0 in the static case. In spherical symmetry:

```
d/dr [r^2 rho v] = 0  =>  rho(r) proportional to 1/r^2.
```

Since amplitude A is proportional to sqrt(rho), we obtain

```
A(r) proportional to 1/r.
```

This inverse-distance scaling is the key ingredient in the force law derivation.

3.3 Force from Coherence Gradient

A test mass M_2 placed at distance r from M_1 interacts with the instruction carrier field through the disformal coupling established in the SCT action (Paper 14, Section 2.3) [2]. In that coupling, the effective potential experienced by M_2 scales as

```
U proportional to M_2 |Psi|^2,
```

and the force is F = -nabla U.

3.4 Stationary-Phase Argument and Dominant-Source Approximation

The total field at x from a continuous source distribution is

```
Psi(x) = integral [rho_s(r-prime) / |x - r-prime|] exp(ik|x - r-
                        prime|) d^3r-prime.
```

Validity condition. The stationary-phase approximation is valid when k * lambda_source >> 1, where lambda_source is the typical inter-source separation scale. We assume throughout that the instruction carrier wavelength lambda = 2*pi/k satisfies lambda << d_nearest, where d_nearest is the distance to the nearest discrete mass. Under this condition the exponential oscillates many times across the source distribution, and contributions from all sources except the nearest are suppressed by destructive interference.

Continuum caveat. In a strictly continuous mass distribution there is no discrete nearest source, and the stationary-phase argument as stated does not apply directly. In practice, all astrophysically relevant situations involve discrete matter concentrations (stars, gas clouds, galaxies) well-separated on the scale of lambda. The continuum description is a convenient integral notation for this discrete physical reality. Situations where sources are truly continuously distributed at sub-wavelength separations require the full integral and are beyond the scope of this paper.

For a discrete point source M_1 at position r_0 = r from the origin, the integral yields:

```
|Psi(x)|^2 approximately equals M_1^2 / r^2 + (subleading terms from
                          other sources).
```

The subleading contributions are suppressed by (M_other/M_1)(r/r_other) relative to the dominant term, as derived in Appendix A.

3.5 Coherence Factor and Baryon-to-Photon Energy Density Ratio

Definition of R_b. We define R_b as the dimensionless ratio of baryon energy density to photon energy density evaluated at matter-radiation equality:

```
R_b = rho_baryon / rho_photon evaluated at (z_eq approximately 3400)
                      approximately equals 0.26.
```

This is *not* the same as the baryon-to-photon number ratio eta approximately equals 6.1 x 10^{-10}. The two are related by a factor that accounts for the baryon rest-mass energy per photon energy at equality. At matter-radiation equality (T_eq approximately equals 0.75 eV), baryons are non-relativistic with energy density rho_baryon approximately equals n_b m_p

c^2 while photons have rho_photon approximately equals n_gamma <E_gamma> approximately equals n_gamma * 2.7 k_B T_eq. The resulting ratio evaluates to R_b approximately equals 0.26. The detailed calculation, including the numerical factor arising from the collision-cascade geometry of the coherence background, is completed in Paper 15 [3], Section 3.1. It cannot be derived from eta alone by the simple formula eta/3.8 without the accompanying energy-scale conversion; that shorthand notation, which appeared in earlier drafts, has been removed.

The coherence correction. Beyond the single-source term M_1^2/r^2, the cosmological background of instruction carriers contributes an additional incoherent baseline to the denominator Sum|A_i|^2 of the coherence function. Evaluating this background contribution (Paper 15 [3], Section 3) introduces the factor $R_b/3$, where 1/3 arises from angular averaging of isotropic background flux relative to the radial source flux (Paper 15 [3], Section 3.2). The effective coherent intensity at r then becomes

```
|Psi|_eff^2 = (1 + R_b/3) * M_1^2/r^2 approximately equals 1.087 *
                            M_1^2/r^2.
```

Note on logical status. The value R_b approximately equals 0.26 and the resulting C approximately equals 1.087 are derived quantities in [3] from the SCT framework, not fitted free parameters. However, a potential circularity must be flagged: R_b approximately equals 0.26 numerically agrees with the standard cosmological baryon-to-photon energy density ratio at matter-radiation equality (Omega_b/Omega_m times rho_matter / (f_gamma rho_radiation) approximately equals 0.16/0.6 approximately equals 0.27, consistent with Planck 2018 parameters). For this to be a genuine prediction rather than a circular input, Paper 15 [3] must derive R_b from SCT collision-cascade parameters alone, without using the measured baryon fraction as an input, and then demonstrate agreement with observation as an independent check. This verification is flagged as an open requirement: if Paper 15's derivation uses the observed baryon fraction as a boundary condition, the agreement is not a cross-check but a tautology. Readers should consult Paper 15 [3] directly to assess this.

3.6 Effective Force Law

Unit analysis and coupling constant. The force on test mass M_2 is

```
F = -G_eff M_2 nabla|Psi|^2.
```

The starting point F = -G_eff M_2 nabla|Psi_source|^2 is not independently derived in this paper; it is the disformal matter coupling from the SCT action (Paper 14 [2], Eq. 23), adopted here as input. The coupling ansatz U proportional to M_2 |Psi_source|^2 is the operative assumption of this framework. With |Psi_source|^2 carrying units [mass^2] (as Psi_source = integral rho_s/r exp(ikr) d^3r-prime carries units of mass), and nabla|Psi_source|^2 having units [mass^2/length], the coupling constant G_eff must have units [length^4/(mass * time^2)] for F to have dimensions of force. This differs by one factor of mass from the standard Newton constant G_N with units [length^3/(mass * time^2)]. The resolution is definitional, not derivational: G_N is defined by requiring that the force law

reproduce the Newtonian result in the classical limit (C-hat approaches 1), which fixes G_N = G_eff / M_1 using the source linearity of the coupling. This paper does not predict the numerical value of G_N; it inherits G_N from observation and from Paper 14 Section 4 where the explicit connection to microscopic SCT parameters (kappa and the dispersion relation) is worked out. The factor of 2 from differentiating M_1^2/r^2 is similarly absorbed into this definition. The correct final expression is therefore a definition of G_N, not a derivation of it:

Substituting the coherence-corrected intensity and taking the gradient with respect to r:

```
nabla|Psi|_eff^2 = -(2/r^3)(1 + R_b/3) M_1^2 r-hat.
```

The factor of 2 here is an artifact of differentiating M_1^2/r^2; it is absorbed into the definition of G_eff, which is matched to G_N by requiring agreement with Newtonian gravity in the classical (C approaches 1, R_b approaches 0) limit. The final force law is:

```
F = G_N (M_1 M_2 / r^2) (1 + R_b/3).
```

The classical inverse-square law emerges with a coherence correction C = 1 + R_b/3. In the limit R_b approaches 0 (negligible baryon background) the standard Newtonian law is exactly recovered.

3.7 Summary of the Derivation

Step	Key element	Mathematical consequence
1	Isotropic emission	dN/dt = kappa M_1
2	Conserved radial flux	A(r) proportional to 1/r
3	Disformal coupling (Paper 14)	F proportional to -nabla|Psi|^2
4	Stationary-phase dominance (k*lambda << d_nearest)	|Psi|^2 approximately M_1^2/r^2
5	Coherence correction (Paper 15)	C = 1 + R_b/3 approximately 1.087
6	G_N matching (Paper 14)	F = G_N M_1 M_2/r^2 * (1 + R_b/3)

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4. Constructive Interference as Dark Matter

We demonstrate that the coherence function C(x,t) naturally produces enhanced gravitational effects that mimic the observational signatures conventionally attributed to dark matter. In this framework, what is interpreted as missing mass arises instead from constructive interference among instruction carriers, leading to excess gravitational acceleration without additional matter.

4.1 Coherence-Enhanced Gravity

From the derivation in Section 3, the effective gravitational force is modulated by the coherence function:

$$F = -G_N (M_1 M_2 / r^2) C(x) \text{ r-hat}.$$

In regions where C > 1, gravitational attraction exceeds the Newtonian prediction for the visible mass alone. The maximum coherence factor in fully virialized halos is A* = 5.970 (derived in Paper 15 [3], Section 3.1, from the collision-cascade geometry of the coherence background). The background cosmological value $C_{bg} = 1 + R_b/3$ approximately equals 1.087 represents a small but in-principle measurable deviation from Newtonian gravity in the weak-field limit.

4.2 Relation to Dark Matter Phenomenology

The conventional dark matter hypothesis posits a non-luminous, collisionless mass component. In the coherence framework, enhanced gravitational acceleration arises from the coherent intensity $|\Psi|^2$ exceeding the incoherent baseline. For a spiral galaxy with visible mass M_{vis}, the observed rotation velocity satisfies

$$v^2(r)/r = G_N M_{vis}(r)/r^2 * C(r).$$

Flat rotation curves require C-hat(r) proportional to r at large radii for a given $M_{vis}(r)$ profile. The background cosmological value $\hat{C}_{bg}$ approximately equals 1.087 is a constant and cannot by itself produce flat rotation curves. It merely rescales the Keplerian curve by 8.7%, leaving $v^2(r)$ proportional to 1/r at large r. Flat curves require a spatially varying C-hat(r) that grows with r, reaching factors of 5 to 10 in the outer halo. The coherence framework provides this through the psi-field equations of Paper 14 (Theorem 4.3) [2], which predict C-hat(r) growing from near-Newtonian values in the inner galaxy to the maximum A* = 5.970 in virialized halos. The background value $\hat{C}_{bg}$ is the cosmological floor; the actual halo profile is determined by the local coherence field solution. This spatial profile introduces the scale parameters r_c and C_0 discussed in Section 4.3.

4.3 Comparison to NFW Profiles

The Navarro-Frenk-White (NFW) profile provides an empirical fit to dark matter halos:

$$\rho_{NFW}(r) = \rho_0 / [(r/r_s)(1 + r/r_s)^2],$$

yielding cumulative mass $M_{NFW}(r)$ proportional to $\ln(1 + r/r_s) - (r/r_s)/(1 + r/r_s)$. In the coherence framework the effective mass is $M_{eff}(r) = M_{vis}(r)\, C(r)$. A simple parameterization consistent with the coherence field equations is

$$C(r) = 1 + C_0 [1 - 1/(1 + (r/r_c)^2)],$$

where C_0 is the maximum enhancement and r_c the coherence scale. The key distinction is ontological: NFW describes a postulated mass distribution, while C(r) describes a coherence-induced modification of the gravitational coupling.

4.4 Comparison to MOND Scaling

MOND posits a modification of Newtonian gravity below an acceleration scale a_0 approximately equals 1.2×10^{-10} m s^{-2}. In the coherence framework the effective acceleration for a point mass M is

$$a(r) = (G_N M / r^2) C(r).$$

To match MOND at large r requires C(r) proportional to r, consistent with the flat rotation curve condition. The conceptual distinction is significant: MOND modifies the law of inertia at a fundamental level, while the coherence framework retains Newtonian dynamics and modifies the effective coupling via the field C(x).

4.5 Comparison Table

Feature	Dark matter (NFW)	MOND	Coherence framework
Enhancement mechanism	Additional mass	Modified dynamics	Coherence factor C > 1
Free parameters	r_s, ρ_0	a_0	xi (from R_b); plus r_c, C_0 for spatial C-hat(r) profile
R_b prediction	None	None	R_b approximately 0.26 (Paper 15)
Observational test	Lensing + dynamics	Rotation curves	Coherence corrections (Paper 16)

4.6 Roadmap to a Unique C-hat(r) Profile

The coherence framework is not yet fully predictive at the galactic level. Section 4.3 introduces the parameterized profile $\hat{C}(r) = 1 + C_0[1 - 1/(1 + (r/r_c)^2)]$ as a phenomenological ansatz; Section 7.2 acknowledges that whether r_c and C_0 are uniquely determined by the baryonic mass distribution is an open question. This section outlines the route to a unique derivation, which constitutes the primary open theoretical problem of the framework.

Step 1: Field equation for Psi_source. The carrier phase field psi obeys the SCT field equation derived in Paper 14 [2] (Theorem 3.1). For a given baryonic mass distribution $\rho_b(r)$, this equation determines Psi_source(x) uniquely as its Green's function solution, analogous to how the Poisson equation uniquely determines the Newtonian potential from a mass distribution. The boundary condition is that Psi_source approaches 0 at spatial infinity.

Step 2: Evaluate I_{bg} from the cosmological background. The background intensity $I_{bg}(x)$ is determined by integrating the carrier flux from all matter in the universe excluding the local source. For an isolated galaxy in the cosmic web, I_{bg} is set by the large-scale matter distribution and varies slowly on galactic scales. In the cosmological limit I_{bg} reduces to the uniform value used to derive $\hat{C}_{bg} = 1 + R_b/3$ in Paper 15 [3]. Corrections

from the local environment (nearby galaxies, the cluster potential) are computable but require knowledge of the surrounding mass distribution.

Step 3: Compute C-hat(r) = 1 + |Psi_source(r)|^2 / I_bg(r). Once Steps 1 and 2 are completed, C-hat(r) follows with no free parameters beyond xi (set by R_b). The profile is uniquely determined by rho_b(r) through the Green's function of the SCT field equation. The parameters r_c and C_0 of the phenomenological ansatz in Section 4.3 would then be outputs of this calculation, not inputs. Whether the resulting C-hat(r) reproduces the observed rotation curves of a representative galaxy sample is the central quantitative test of the framework.

Current status. Steps 1 and 2 require numerical solutions of the SCT field equations for realistic galactic mass distributions. These solutions are not completed in the present series of papers. Paper 14 [2] establishes the field equations and proves existence and uniqueness of solutions for the NIPOK metric (Theorem 6.1); extending this to the non-stationary, non-axisymmetric case of a general galactic mass distribution is the subject of ongoing work. Until this is complete, the framework is predictive at the cosmological level (where C-hat_bg is analytically derived) but remains parametric at the galactic level (where the C-hat(r) profile requires the numerical solutions). The baryonic Tully-Fisher relation (v^4 proportional to M_vis) and the mass discrepancy-acceleration relation are the key observational benchmarks that the completed calculation must reproduce.

---

5. Modified Einstein Equations

We incorporate the coherence function C(x,t) into the gravitational field equations. The effective stress-energy tensor that sources curvature is modulated by the coherence field.

5.1 Coherence-Modified Stress-Energy

The SCT-MASTER equation, derived from the SCT action in Paper 14 (Theorem 3.1) [2], is:

```
G_{mu nu} + Lambda_eff(x,t) g_{mu nu} = (8*pi*G_N/c^4) [T_{mu
              nu}^(matter) + T_{mu nu}^(sup)(A)],
```

where T^(sup)_{mu nu}(A) = (A - 1) T^(baryon)_{mu nu} represents the superposition stress-energy from coherent structures. In the cosmological limit where A = C and Lambda_eff approaches 0 for virialized regions, this reduces to

```
G_{mu nu} = (8*pi*G_N/c^4) C(x,t) T_{mu nu}^(matter).
```

5.2 Conservation Laws and the Bianchi Condition

The Bianchi identities impose nabla^mu G_{mu nu} = 0. For consistency with the modified equations:

$$\nabla^\mu [C(x,t)\, T_{\mu\nu}^{(matter)}] = 0.$$

Expanding, and using $\nabla^\mu T_{\mu\nu}^{(matter)} = 0$ for matter in the absence of non-gravitational forces:

$$T_{\mu\nu}^{(matter)} \nabla^\mu C = 0.$$

For pressureless dust, $T_{\mu\nu} = \rho u_\mu u_\nu$, this gives $u^\mu \partial_\mu C = 0$: the coherence function must be constant along matter worldlines.

Status of the reduced field equation. The equation $G_{\mu\nu} = (8\pi G_N/c^4)\, \hat{C}\, T_{\mu\nu}^{(matter)}$ presented in this paper is an approximation, not an exact field equation. The Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ is a mathematical identity, so the right-hand side must satisfy $\nabla^\mu(\hat{C}\, T_{\mu\nu}) = 0$ exactly for this equation to be consistent. This requires $u^\mu \partial_\mu \hat{C} = 0$: $\hat{C}$ constant along matter worldlines. In general this is not satisfied, and the reduced equation is inconsistent at the level of $|\Delta \hat{C}/\hat{C}|$. The exact field equation. which is mathematically consistent. is the SCT-MASTER equation in Paper 14 [2], from which the reduced form here is obtained by setting Λ_{eff} approaches 0 and neglecting the psi-field kinetic terms. The present paper uses the reduced equation as a working approximation valid when $|u^\mu \partial_\mu \hat{C}| / (\hat{C} H_0) \ll 1$. Quantitative estimate:

This condition is approximately satisfied when the fractional change in C over one dynamical time $\tau_{dyn} = (G_N \rho)^{-1/2}$ is small: $|\Delta C / C| \ll 1$ per τ_{dyn}. For a galaxy with $\rho \sim 10^{-24}$ g cm^{-3} and $\tau_{dyn} \sim 10^8$ yr, the coherence field varies on Hubble timescales $H_0^{-1} \sim 1.4 \times 10^{10}$ yr, so $|\Delta C / C|_{\tau_{dyn}} \sim \tau_{dyn} / H_0^{-1} \sim 10^{-2}$. The Bianchi consistency condition is thus satisfied to 1% accuracy for galactic systems, which is well within observational uncertainty. For cluster and cosmological scales the same ratio applies with the appropriate dynamical time. The full Noether identity from diffeomorphism invariance of S_{SCT}, which shows that exact conservation holds when the complete SCT action including the psi-field kinetic term is used, is derived in Paper 14 [2].

5.3 Effective Growth Suppression

The matter density contrast $\delta = \delta\rho/\bar{\rho}$ obeys

$$\ddot{\delta} + 2H \dot{\delta} = 4\pi G_N C_{bg} \bar{\rho}\, \delta,$$

in the Newtonian limit. Spatial variations in C introduce a scale-dependent suppression: for perturbations with wavenumber $k > 1/\xi$, coherence is incomplete and the effective gravitational coupling is reduced. This suppresses structure formation on small scales without requiring baryonic feedback.

5.4 Summary of Field Equations

Equation	Standard GR	Coherence-modified
Einstein	$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$	$G_{\mu\nu} + \Lambda_{eff}\, g_{\mu\nu} = 8\pi G(T + T^{sup})$ (Paper 14)

Background Friedmann	H^2 = (8*pi*G/3) rho-bar	H^2 = (8*pi*G/3) C_bg rho-bar
Growth equation	delta-double-dot + 2H delta-dot = 4*pi*G rho-bar delta	delta-double-dot + 2H delta-dot = 4*pi*G C_bg rho-bar delta
Small-scale suppression	None	delta C < 0 for k > 1/xi

---

6. Testable Predictions

6.1 Galaxy Rotation Curves

The coherence factor C = 1 + R_b/3 approximately equals 1.087 predicts a specific amplitude of gravitational enhancement. For galaxies, rotation velocities are

```
v^2(r) = G_N M_vis(r)/r * C(r),
```

yielding an asymptotic velocity v_flat. Note that a constant C-hat_bg approximately equals 1.087 alone does not produce a flat rotation curve: it gives v^2(r) = G_N M_vis(r-approaching-infinity) C-hat_bg / r, which is a Keplerian decline (v^2 proportional to 1/r) rescaled by 8.7%. Flat rotation curves require the spatially growing C-hat(r) from the coherence field solution of Paper 14 (Theorem 4.3) [2], which satisfies C-hat(r) proportional to r at large radii. The asymptotic circular velocity is set by the intersection of the baryonic mass profile with the coherence field profile, both of which are determined by R_b.

6.2 Gravitational Lensing Anomalies

The lensing convergence becomes

```
kappa_eff(theta) = (4*pi*G_N/c^2) integral dz W(z) C(x(z))
                   rho_baryon(x(z)),
```

predicting coherence peaks at sub-kpc scales where constructive interference yields C > 1.1, detectable with ALMA or ngEHT.

6.3 Scale-Dependent Structure Suppression

The finite coherence length xi introduces a characteristic suppression scale

```
k_c ~ 1/xi approximately equals 0.5 h Mpc^{-1},
```

derived in [3] from the baryon density and coherence relaxation dynamics. This predicts reduced satellite galaxy counts and core-like dark matter profiles, measurable with BOSS, DESI, and Euclid.

6.4 CMB Consistency

The coherence framework yields CMB power spectra identical to LCDM for l < 2000, with deviations appearing at l > 2000 from coherence suppression. Baryon acoustic oscillations are unaffected at large scales, preserving the standard BAO ruler.

6.5 Summary of Predictions

Prediction	Observable	Scale	Discriminator
Rotation amplitude	v_flat set by C-hat(r) proportional to r profile (Paper 14 Thm 4.3); C-hat_bg approximately 1.087 is floor only	kpc to Mpc	C_bg approximately 1.087
Lensing peaks	Localized C > 1.1	sub-kpc	No dark matter equivalent
Small-scale suppression	k_c approximately 0.5 h Mpc^{-1}	10 to 100 Mpc	Cutoff in matter spectrum
CMB deviations	l > 2000	Cosmological	Suppression of power

---

7. Comparison to LCDM

7.1 Fundamental vs. Emergent

LCDM treats gravity as a fundamental interaction. Dark matter is a postulated additional particle species; dark energy is introduced as a cosmological constant or dynamical field. The coherence framework derives gravitational phenomenology from the collective interference of instruction carriers; all gravitational effects follow from the coherence properties of the electromagnetic substrate, as established in Papers 14 and 15 [2, 3].

7.2 Parametric Complexity

The base LCDM model requires 6 parameters (Omega_b h^2, Omega_c h^2, Omega_Lambda, H_0, n_s, A_s). With extensions (neutrino masses, dark energy equation of state, inflation parameters) the count grows to 10 to 20. At the cosmological level, the coherence framework introduces one new quantity, the coherence length xi, whose value is determined by R_b as derived in [3]. The absence of a dark matter density parameter Omega_c reduces the standard count by one. At the galactic level, flat rotation curves require solving the coherence field equation (Paper 14, Theorem 4.3) [2] for each system; the solution introduces a spatial profile C-hat(r) with scale parameters r_c and C_0 (Section 4.3). These are analogous to the NFW parameters r_s and rho_0, though in the coherence framework they are in principle determined by the baryonic mass distribution via the field equations rather than fitted independently. Whether this determination is unique and

computable without additional free parameters is an open question pending full numerical solutions of the coherence field equations.

7.3 Particle Dark Matter vs. Wave Interference

Dark matter is assumed to be a particle species. Despite extensive experimental effort, no such particle has been detected. The coherence framework replaces the particle with constructive interference of instruction carriers, naturally yielding cored profiles and scale-dependent suppression of structure without additional feedback mechanisms.

7.4 Relation to Existing Modified Gravity Literature

The coherence framework shares structural features with several existing proposals and must be distinguished from them. Verlinde (2011, 2016) derives an entropic emergent gravity in which gravitational dynamics arise from an elastic response of an underlying information medium; the coherence framework shares the emergent character of gravity but derives from an electromagnetic substrate rather than an entropic one, and makes different predictions for galaxy cluster dynamics. Moffat's MOG (Modified Gravity, Moffat 2006) and Tensor-Vector-Scalar (TeVeS, Bekenstein 2004) theories modify gravity through additional vector and scalar fields; the coherence framework similarly introduces a new field (the instruction carrier field psi) but attributes its origin to electromagnetic rather than gravitational sector physics. On the dark matter side, ultralight (fuzzy) dark matter models (Hui et al. 2017) explain small-scale structure suppression through de Broglie wavelength effects of a ~10^{-22} eV boson; the coherence framework produces analogous scale-dependent suppression through the finite coherence length xi, with the physics being wave interference rather than quantum pressure. MOND (Milgrom 1983; Famaey and McGaugh 2012) successfully predicts the baryonic Tully-Fisher relation from a universal acceleration scale a_0; the coherence framework must reproduce this relation from its field equations to be considered viable, and this has not yet been demonstrated explicitly. These comparisons identify four open tests that Paper 16 [1] and future work must address.

7.5 Summary

Aspect	LCDM	Coherence framework
Status of gravity	Fundamental	Emergent
Dark matter	Postulated particle	Constructive interference
Free parameters	6 (base) to approximately 20 (with extensions)	1 (xi, determined by R_b) at cosmological level; r_c, C_0 for galactic C-hat(r) profile (open question. Section 7.2)
Small-scale structure	Requires baryonic feedback	Naturally suppressed by xi
Coherence factor	Not present	C = 1 + R_b/3 approximately 1.087 (Paper 15)

Observational confirmation	Extensive, independent	Paper 16 [1] (self-citation; independent replication pending)

---

8. Robustness to Systematics

8.1 Intrinsic Alignments

Intrinsic alignment (IA) refers to correlations between galaxy shapes caused by local tidal fields. In the coherence framework, the gravitational potential is sourced by C(x) rho_baryon(x), where C is a geometric field determined by the instruction carrier distribution. Because IA arises from the same tidal field causing lensing, and because the coherence modification applies universally to the potential, the relative IA contribution is unchanged from the Newtonian expectation.

8.2 Photometric Redshift Uncertainties

The lensing convergence integrates C(x(z)) rho_baryon(x(z)) along the line of sight, weighted by the lensing efficiency kernel W(z). Since C varies on the coherence length scale xi >> the typical photometric redshift scatter sigma_z c/H_0, the integrated signal is insensitive to small-scale redshift errors.

8.3 Baryonic Feedback

There is no dark matter component to separate from baryons. The gravitational field is sourced directly by baryons, modulated by C. Baryonic feedback affects the baryon distribution itself, which is directly observable. The coherence factor C applies uniformly; no additional feedback parameters are required.

8.4 Summary

Systematic	LCDM approach	Coherence framework
Intrinsic alignments	Empirical model with A_IA	Geometric prediction from C
Photo-z errors	Spectroscopic calibration required	Insensitive (C varies on scale xi)
Baryonic feedback	Simulation-based marginalization	Direct baryon census suffices

---

9. Bayesian Perspective

9.1 Parameter Space Reduction

Standard dark matter analyses include: Omega_c h^2, halo concentration c_vir, baryonic feedback parameters (A_AGN, A_SN), intrinsic alignment amplitude A_IA, and neutrino mass Sum m_nu. In the coherence framework, R_b approximately equals 0.26 is derived from the SCT action in Paper 15 [3] and corroborated by the observed baryon fraction at matter-radiation equality. It is not a fitted parameter. The coherence length xi enters as the sole new parameter, but its value is fixed by R_b via xi = sqrt(D/Gamma_c) (Section 2.5), reducing the effective free-parameter count relative to the standard model.

9.2 Bayesian Evidence

The Bayesian evidence for model M given data D is

```
Z = integral L(D|theta, M) pi(theta|M) d theta.
```

Paper 16 [1] reports a log-evidence difference of ln B approximately equals 3.78 (B approximately equals 44) in favor of the coherence framework over LCDM with dark matter, using galaxy rotation curves, gravitational lensing, and large-scale structure data. **Caveat.** This result is currently based on analyses by the same research group that developed the framework (Papers 14 to 17), which constitutes a closed self-citation loop within a single series. The Bayes factor B approximately equals 44:1 (Paper 16 [1]; self-analysis) is presented as a provisional, internally consistent result, not as an independently confirmed finding. External replication using Euclid Year 1 data (expected 2027) and independent codes is required for confirmation. Readers should weight this result accordingly.

9.3 Summary

Aspect	LCDM + dark matter	Coherence framework
DM parameters	3 to 5 (fitted)	0
Baryonic feedback	1 to 3 (marginalized)	0
R_b status	Omega_c fitted	Derived (Paper 15), self-cited
Bayes factor vs. LCDM	Reference	B approximately 44:1 (Paper 16 [1]; self-analysis; Euclid Y1 2027 replication required)

---

10. Illustrative Results

10.1 Derivation of the Inverse-Square Law

The derivation begins with a point mass M_1 at the origin emitting instruction carriers isotropically. Radial flux conservation requires rho proportional to 1/r^2 and A proportional to 1/r. At distance r, a test mass M_2 experiences a force proportional to -nabla|Psi|^2. The stationary-phase argument (valid for k*lambda << d_nearest) shows that |Psi|^2

approximately equals M_1^2/r^2. The cosmological background introduces the correction C = 1 + R_b/3 approximately equals 1.087 (Paper 15). After matching units via G_N (Paper 14), the final force law is F = G_N M_1 M_2/r^2 * (1 + R_b/3). The $1/r^2$ law is not assumed; it follows from flux conservation.

10.2 Coherence Function for Overlapping Spheres

Where two mass-induced spheres of influence overlap, constructive interference produces coherence peaks reaching C approximately equals 1.12, with spatial extent approximately 500 pc for xi approximately 1 kpc. These peaks produce lensing convergence anomalies kappa_peak approximately equals 1.12 times kappa_baryon with no corresponding stellar or gas emission. This is a discriminating prediction.

10.3 Galaxy Rotation Curve Comparison

For the Milky Way, using the observed baryonic mass distribution and xi approximately equals 10 kpc, the predicted rotation curve rises in the inner disk and flattens at v approximately equals 220 km s^{-1} in the mid-disk, consistent with observations. An NFW halo with c = 12 and M_vir = 1.5 x 10^{12} M_sun [4] matches comparably but requires an additional fitted parameter (M_vir). The coherence framework uses xi approximately equals 10 kpc (derived from R_b via [3]) as the only cosmological input. Whether the C-hat(r) profile for the Milky Way is then uniquely determined by the baryonic mass distribution without additional free parameters is an open question (Section 7.2); the rotation curve shown here is illustrative of the framework's qualitative behavior, not a parameter-free prediction.

10.4 Sub-kiloparsec Coherence Peaks

At rho_baryon ~ 10^{-24} g cm^{-3}, the coherence length is xi ~ 1 kpc. Predicted peaks have size approximately 500 pc and amplitude C approximately equals 1.12. The corresponding lensing signal is within the detection threshold of ALMA and ngEHT. Detection of such peaks with no correlated stellar or gas emission would strongly discriminate the coherence framework from particle dark matter models.

---

11. Conclusion

We have presented the foundational derivation of the inverse-square law from conserved radial flux and the stationary-phase dominance of the nearest source. The coherence factor C = 1 + R_b/3 approximately equals 1.087 links gravitational dynamics to the baryon-to-photon energy density ratio at matter-radiation equality; its derivation from the SCT collision-cascade geometry is the content of [3], Section 3.1. The SCT-MASTER equation, derived variationally in Paper 14 [2], provides the field-equation framework in which C enters as a multiplicative modulation of the matter stress-energy tensor.

The classical limit (C approaches 1) recovers standard Newtonian and GR physics exactly. The coherence correction C approximately equals 1.087, the maximum amplification A* = 5.970 (Paper 15), and the suppression scale k_c approximately equals 0.5 h Mpc^{-1} are all derived quantities, not free parameters. Provisional Bayesian evidence from Paper 16 [1] favors the framework by B approximately equals 44 over LCDM with dark matter; independent replication by external groups is the necessary next step.

On the interpretation of gravitational attraction: the force is real; it emerges from the gradient of coherent intensity in the instruction carrier field. The Newtonian 1/r^2 law is a consequence of flux conservation, not a fundamental postulate. Dark matter is re-interpreted as a misidentification of coherence enhancement. At the cosmological level the framework introduces one new parameter (xi) determined by R_b. At the galactic level, whether the spatial C-hat(r) profile is uniquely fixed by the baryonic mass distribution without additional free parameters remains an open question (Section 4.6); numerical solutions of the SCT field equations for realistic galactic profiles are required. Two further open problems define the boundary of what has been established in this series: (1) G_N is matched to observation rather than derived from SCT parameters. Its first-principles derivation is the content of Paper 14 Section 4 and is not completed here. (2) The electromagnetic sector (epsilon_0, mu_0) requires completing two open analyses of the SCT field equations: first, whether the condensate's Goldstone mode produces a massless spin-1 excitation. which would force local U(1) invariance as a consistency requirement rather than a postulate (Weinberg 1964). and second, whether the field equations admit topologically stable vortex solutions with quantized winding number, which would ground the U(1) bundle structure in the topology of the carrier condensate, analogous to the Abrikosov vortex structure of type-II superconductors. Neither analysis has been carried out; they constitute well-defined mathematical questions about Paper 14's field equations, not open-ended conceptual problems. The framework is predictive and internally consistent at the cosmological level; the galactic and gauge-sector open questions are clearly identified and tractable.

Gravity is not the curvature of spacetime. Gravity is the geometry of coherence.

Bonus Exercise: Roadmap to Deriving epsilon_0 and mu_0. Open Questions in the Gauge Sector

The framework presented here establishes gravity as an emergent phenomenon arising from coherence in an electromagnetic substrate. A natural extension is whether the same substrate can recover the constants of electromagnetism from first principles. The following is a roadmap, not a completed derivation.

Step 1: Postulate a Lagrangian for psi

Begin with a Lagrangian density for the instruction carrier field with a symmetry-breaking potential:

```
L = partial_mu psi* partial^mu psi - mu^2|psi|^2 - lambda|psi|^4.
```

For mu^2 < 0 the symmetric state <psi> = 0 is unstable; the system condenses to a coherent ground state with |psi_0|^2 = -mu^2/(2*lambda).

Step 2: Identify the Order Parameter

The coherence function C = |<psi>|^2/<|psi|^2> emerges as the normalized condensate density. Expand around the coherent background: psi(x,t) = psi_0 + delta psi(x,t), where |psi_0|^2 is set by R_b ([3]).

Step 3: Derive the Dispersion Relation and Speed of Light

Linearized equations for delta psi yield a mode spectrum. When symmetry is broken, a massless Goldstone mode appears. Its phase velocity defines c:

```
c = 1 / sqrt(|psi_0|^2 * kappa),
```

with kappa a stiffness parameter from the Lagrangian. This step requires identifying the appropriate dispersion relation for the specific potential chosen.

Step 4: The Status of U(1) Gauge Invariance

Write psi = psi_0 exp(i*theta). Integrating out the massive amplitude fluctuations delta|psi| leaves an effective action for the Goldstone phase theta with a global U(1) symmetry theta approaches theta + const. The question is whether local gauge invariance. theta(x) approaches theta(x) + chi(x) for arbitrary chi(x). must be separately postulated or whether it can be derived.

Route 1: Consistency of a massless spin-1 field (established result). If the condensate's Goldstone mode couples to a massless spin-1 excitation, local gauge invariance is not a free postulate. It is forced by the consistency requirements of the interacting field theory. A massless spin-1 particle has two physical polarizations. A free massive spin-1 field has three. For a massless field, the third polarization is unphysical and must be removed; in an interacting Lorentz-invariant theory, the only consistent way to do this is through a gauge symmetry (Weinberg 1964, Phys. Rev. 135, B1049; Wald 1986, GRG 18, 159). This result, established in standard quantum field theory, means the postulate is not "local U(1) invariance" but rather "the Goldstone mode of the broken SCT condensate produces a massless spin-1 excitation that couples to charged matter." This is a claim about the field spectrum of the SCT Lagrangian that is in principle checkable from Paper 14's dispersion relation analysis. If that spectrum contains a massless spin-1 mode, local gauge invariance follows without further input. Whether it does is an open question about Paper 14's field equations, not about the gauge principle itself.

Route 2: Topological vortex structure (open question). A deeper route addresses why psi should be treated as a section of a U(1) principal bundle rather than simply a complex scalar field on flat spacetime. This is a question about the global topology of the carrier field configuration space. If the SCT phi-field equations admit stable vortex solutions.

field configurations in which the phase theta winds by 2*pi*n around a topological defect, with the winding number n in Z conserved by the equations of motion. then the configuration space of the theory is classified by pi_1(S^1) = Z, which is precisely the first Chern class of a U(1) bundle. In that case, the U(1) bundle structure is not an input but an output: it is the topological classification of the stable field configurations. This is the mechanism by which U(1) gauge structure arises in the Ginzburg-Landau theory of superconductors (Abrikosov vortices, Nobel Prize 2003) and in superfluid helium (quantized vortices), where the gauge structure emerges from the topology of the order parameter rather than from a postulate. Whether the SCT field equations of Paper 14 [2] support quantized vortex solutions of this kind has not been analyzed and constitutes the correct open question. If such solutions exist, the U(1) gauge structure of the electromagnetic sector would be derived from the topology of the instruction carrier condensate, completing the derivation without additional postulates. If they do not exist, an independent postulate would remain necessary.

Under either route, if gauge invariance is established, minimal coupling partial_mu theta approaches partial_mu theta + q*A_mu then yields the Maxwell action

```
L_EM = -(1/4*mu_0) F_{mu nu} F^{mu nu}.
```

Step 5: Express epsilon_0 and mu_0 in Terms of Coherence Parameters

Reading off coefficients:

```
mu_0 = 1/(q^2 |psi_0|^2),   epsilon_0 = q^2 |psi_0|^2 / c^2.
```

Both constants derive from the instruction carrier condensate density |psi_0|^2. The fine-structure constant alpha = e^2/(4*pi*epsilon_0*hbar*c) then follows from the gauge coupling q and the electron coupling e.

Step 6: Varying Coherence Predicts Vacuum Dispersion

Because |psi_0|^2 proportional to C(x,t), spatial variations in C imply spatial variations in epsilon_0 and mu_0. This predicts measurable vacuum dispersion and birefringence near matter concentrations. Compute the expected magnitude and identify observational constraints.

Step 7: Complete the Derivation

The steps above outline the logical path. Completion requires resolving two specific open questions about the SCT field equations of Paper 14 [2]: (i) whether the linearized spectrum around the condensate ground state contains a massless spin-1 mode (which would force local U(1) invariance via the consistency argument of Weinberg 1964), and (ii) whether the nonlinear field equations admit topologically stable vortex solutions with quantized circulation (which would ground the U(1) bundle structure topologically, as in the Ginzburg-Landau theory of superconductivity). These are well-defined mathematical questions, not programmatic gaps. If both are answered affirmatively, Steps 1 through 7 constitute a complete derivation of epsilon_0 and mu_0 from the SCT condensate without independent postulates.

12. Code and Appendices

12.1 Code: Continuum Coherence Integral

The following pseudo-code implements the coherence function C(x,t) in the continuum limit, following the four-step derivation of Section 2.

```
FUNCTION Compute_Coherence(x_target, source_distribution, k, N_samples)
    source_positions = DISCRETIZE(source_distribution, N_samples)
    source_amplitudes = source_distribution[source_positions] *
VOLUME_ELEMENT
    coherent_sum_real = 0.0
    coherent_sum_imag = 0.0
    incoherent_sum    = 0.0
    FOR each source i IN source_positions
        r   = DISTANCE(x_target, source_positions[i])
        A   = source_amplitudes[i] / r           // A proportional to 1/r
(flux conservation)
        phi = k * r + PHASE_OFFSET(source_positions[i])
        coherent_sum_real = coherent_sum_real + A * COS(phi)
        coherent_sum_imag = coherent_sum_imag + A * SIN(phi)
        incoherent_sum    = incoherent_sum    + A * A
    END FOR
    coherent_intensity = coherent_sum_real^2 + coherent_sum_imag^2
    C = coherent_intensity / incoherent_sum
    RETURN C
END FUNCTION
```

The incoherent_sum in the denominator is the power sum Sum A_i^2, correctly matching the definition in Section 2.1. The stationary-phase refinement identifies the nearest source as dominant and is valid when k * DISTANCE(x_target, nearest) >> 1.

Appendix A: Stationary-Phase Dominance for Discrete Sources

A.1 General Formulation

Consider the integral

```
I(x) = integral_{R^3} [rho_s(r-prime) / |x - r-prime|] exp(ik|x - r-
                                prime|) d^3r-prime,
```

where k is large relative to the inverse inter-source separation: k * d_nearest >> 1. This condition is the validity requirement for the approximation; it states that the carrier wavelength lambda = 2*pi/k is much smaller than the distance to the nearest source. Under this condition the exponential oscillates rapidly as r-prime varies over the source distribution.

A.2 Stationary Points

The gradient of the phase Phi(r-prime) = k|x - r-prime| with respect to r-prime is

```
nabla_{r-prime} Phi = -k (x - r-prime) / |x - r-prime|.
```

This vanishes only at r-prime = x, which is a singular point of the integrand (A proportional to 1/|x - r-prime| diverges there). In the continuum limit, this singularity indicates that the integral is dominated by the region nearest to x, not that a simple stationary-phase expansion applies at r-prime = x. For a discrete source distribution the integral reduces to a finite sum, and the singular point is not in the domain (sources are at fixed positions r_i not equal to x for a test point not coinciding with a source). The stationary-phase terminology then applies in the sense that contributions from sources far from x are suppressed by rapid phase oscillation.

A.3 Leading-Order Approximation for a Discrete Source

For a discrete source M_0 at position r_0 (distance r_0 = |x - r_0| from the test point), the contribution to the field is

```
Psi_0 = (M_0 / r_0) exp(i*k*r_0).
```

For all other sources at distances r_i > r_0, contributions are suppressed relative to this leading term. The suppression mechanism requires finite carrier bandwidth, which is established by the finite coherence time tau_c = 1/Gamma derived in Section 2.3: a finite coherence time implies a spectral width Delta_omega = Gamma in the frequency distribution P(Delta_omega), and via the dispersion relation a corresponding wave-number spread Delta_k = Delta_omega/v_phase. When Delta_k * (r_i - r_0) >> 1, the phase k r_i varies by more than 2*pi across the bandwidth, and contributions from source i average to zero upon integration over the spectrum. The suppression ratio for time-averaged intensity is (M_i/M_0)(r_0/r_i). This argument requires monochromatic carriers to be replaced by spectrally broadened carriers with coherence length l_c = v_phase/Gamma; if l_c << r_1 - r_0 (separation between nearest and next-nearest source), the dominant-source approximation holds. In the astrophysical context l_c is determined by the collision-cascade dynamics in Paper 15 [3] and is much smaller than typical inter-source separations. The leading-order approximation is therefore

```
|Psi(x)|^2 approximately equals M_0^2 / r_0^2 [1 +
              O((M_1/M_0)(r_0/r_1))],
```

where M_1 and r_1 refer to the next-nearest source. The approximation is good when (M_1/M_0)(r_0/r_1) << 1.

A.4 Subleading Contributions and the Coherence Factor

The sign of the coherence correction requires careful explanation, as it may appear paradoxical: why does the presence of background carriers make C-hat > 1 rather than reducing it? The answer lies in the correct reading of the formula. The gravitational coupling enhancement factor is defined as C-hat = (I_bg + |Psi_source|^2) / I_bg = 1 + |Psi_source|^2/I_bg. Here I_bg is the total incoherent background intensity that would be present at the test point even in the absence of the local source M_0. It is the noise floor set by all other matter in the universe. When M_0 is introduced, its coherent field Psi_source deposits additional intensity |Psi_source|^2 = M_0^2/r_0^2 on top of this background. The ratio |Psi_source|^2/I_bg = R_b/3 is evaluated by computing I_bg from the full cosmological background distribution and expressing the source intensity relative to it (Paper 15 [3],

Section 3). The result C-hat = 1 + R_b/3 > 1 is correct because the source field adds coherently above the background: the numerator grows while the denominator (the pre-existing I_bg) is fixed by cosmology, not by the source. Adding more background carriers would indeed increase I_bg and thereby reduce C-hat. But that is a different scenario (increasing the noise floor) from what we are computing (evaluating the signal-to-noise at a fixed cosmological epoch). The subleading contributions from background sources referenced in earlier drafts entered the old denominator Sum|A_i|^2 of the microscopic coherence function c. That denominator grew, reducing c, which is consistent with c <= 1. In the new framework, those same background carriers set I_bg; the source M_0 then sits above this background, giving C-hat = 1 + |Psi_source|^2/I_bg > 1. The two pictures are mutually consistent: c <= 1 always (local phase alignment bounded), C-hat > 1 possible (source stands above the cosmic noise floor). The numerical evaluation of R_b/3 = |Psi_source|^2/I_bg for the cosmological background is the content of Paper 15 [3], Section 3, and the factor 1/3 arises from angular averaging of isotropic background flux relative to radial source flux.

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